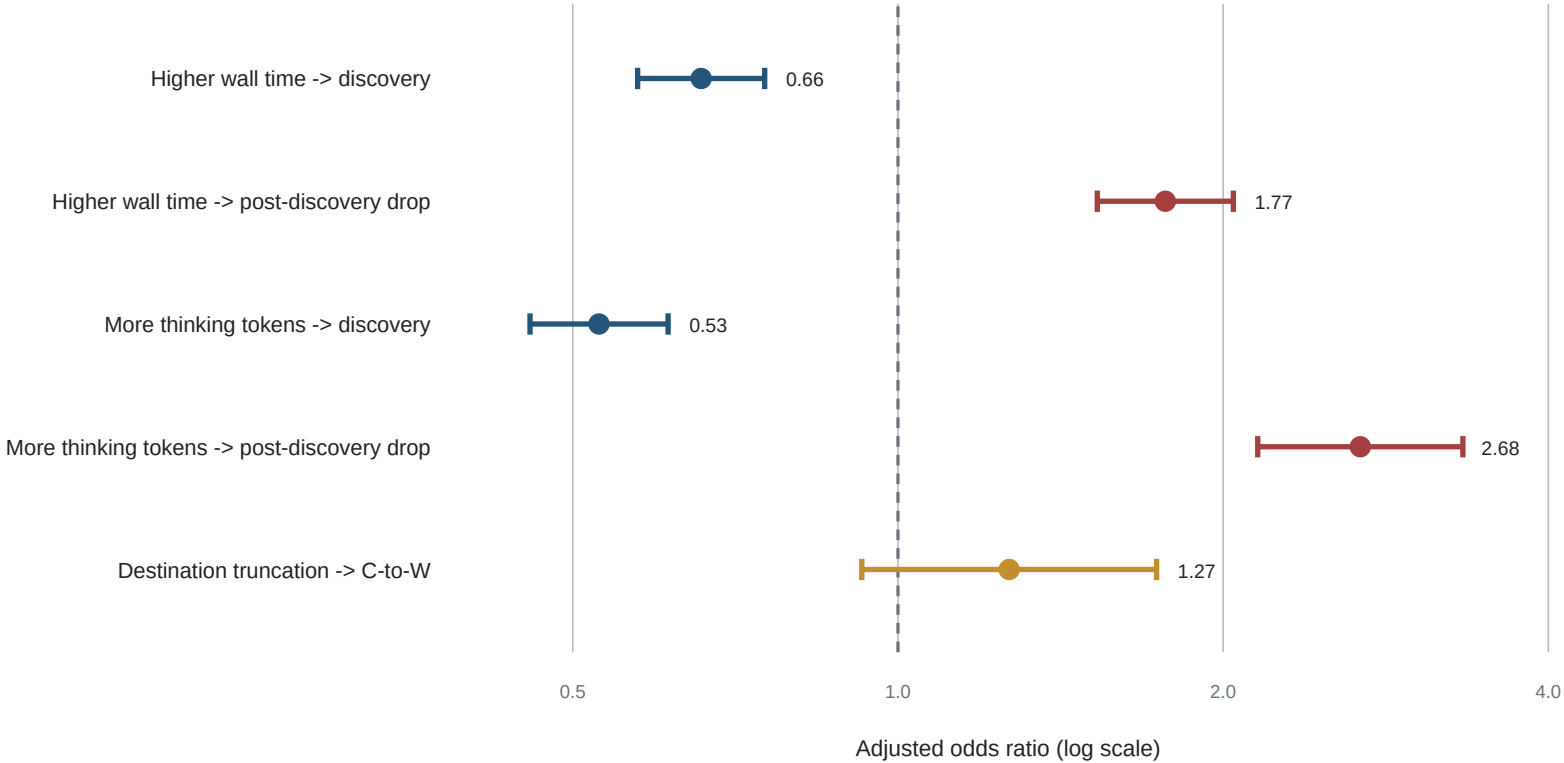Resource exposure and semantic outcomes